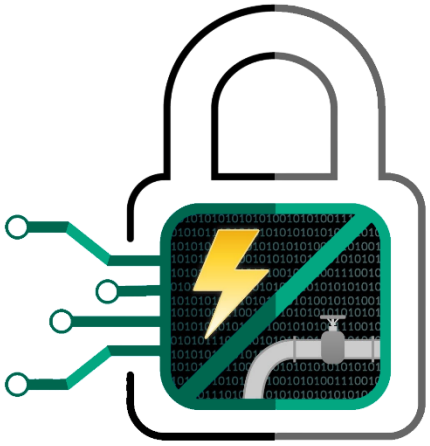


U.S. DEPARTMENT OF ENERGY'S

CYBERFORCE COMPETITION®

DEFENDING U.S. ENERGY INFRASTRUCTURE

ICS Documentation

2026

CYBERFORCE COMPETITION®

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NOTE

Both the HMI and DNP3 PLC VMs are within the Assume Breach Infrastructure and **EXTREME CAUTION** should be used when attempting to alter these machines.

CONNECTIVITY

MODBUS PLC (TRADITIONAL SIDE)

Ubuntu 22.04

plc1.<bluexxxx>.cfc.local
blueteam : BlueTeam2026!

PLC - :502
10.0.x.144:8080
openplc : openplc

DNP3 PLC (ASSUME BREACH SIDE)

Debian 13

plc2.<bluexxxx>.cfc.local
blueteam : `BlueTeam2026!`

PLC - :20000

MQTT -: 1883

CNC/HMI (ASSUME BREACH SIDE)

Windows Server 2019

hmi.<bluexxxx>.cfc.local
blueteam : `BlueTeam2026!`

Ignition Gateway - :8088
blueteam : `BlueTeam2026!`

Data Historian MySQL - 3306
`historian : CyberForce2026!`
Database = historian

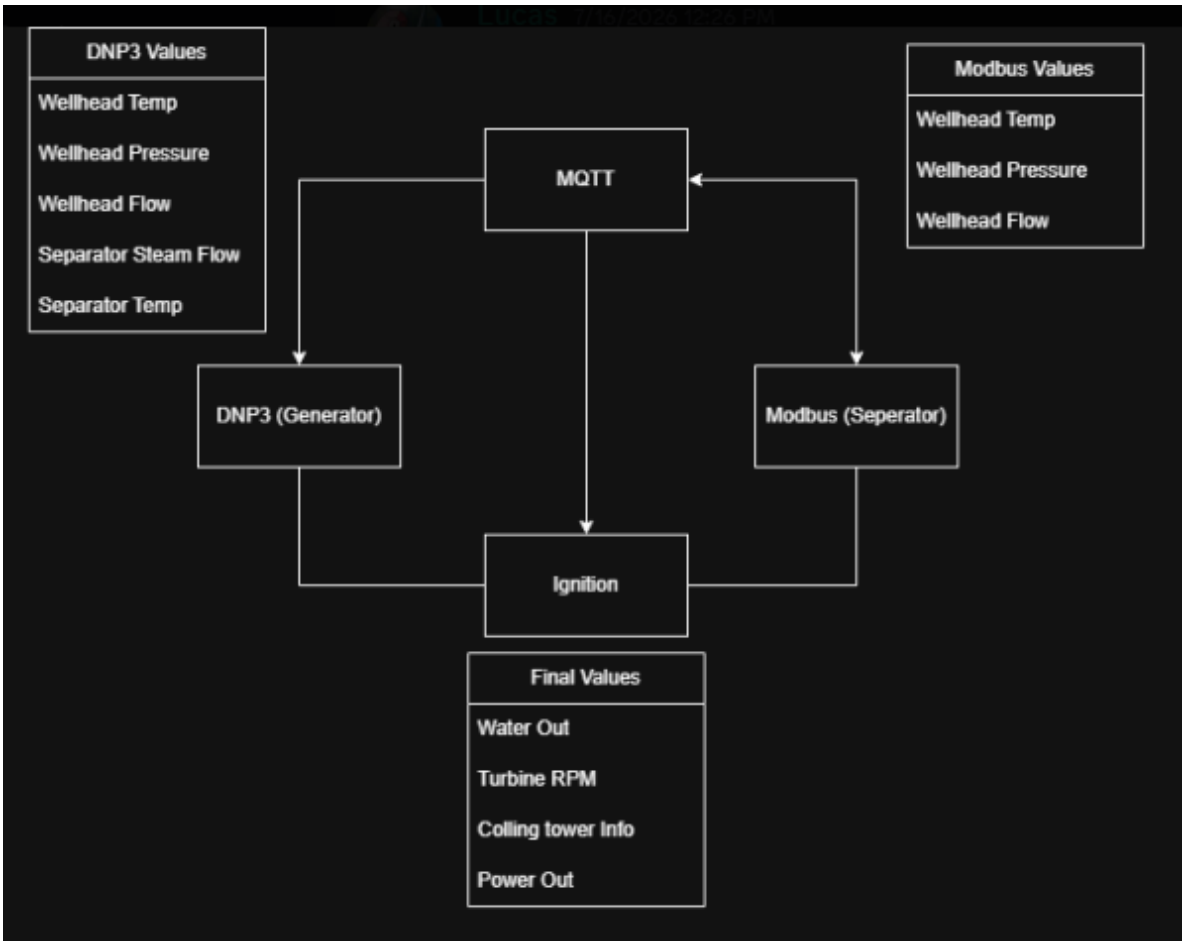
For more information about Ignition please visit:
https://www.inductiveuniversity.com/courses/ignition/ignition-overview/8.1

VULCANA DYNAMICS LIMITED INDUSTRIAL CONTROL SYSTEM MANUAL

SYSTEM OVERVIEW

This year, we are utilizing 4 different components to create our ICS environment. MQTT is a message broker commonly used in Industrial Internet of Things (IIoT) devices to allow seamless communications. This year you will receive all external sensor data from MQTT topics sent from the scoreboard. That data will then be sent to

your 2 PLC devices, Modbus within PLC1 and DNP3 within PLC2. The Modbus device also has the capability to send over its calculated and sensed values required by DNP3 for calculations. Ignition is then used to both visualize the ICS data and reconcile the data between the 3 data sources. Below is an illustration of the pipeline of sensor data and communications.



REGISTERS (2048-2094) AND COILS (0000-0005)

```
VAR
manover AT %QX0.0 : BOOL := FALSE;
ESD AT %QX0.1 : BOOL := FALSE;
StartSystem AT %QX0.2 : BOOL := FALSE;
StopSystem AT %QX0.3 : BOOL := FALSE;
Pressure_Release AT %QX0.4 : BOOL := FALSE;
Separator_Valve_Percent AT %MD0 : REAL := 0.0;
Separator_Pressure AT %MD1 : REAL := 0.0;
Separator_Temp AT %MD2 : REAL := 0.0;
Flash_Factor AT %MD3 : REAL := 0.0;
Sep_Press_Optimal AT %MD4 : REAL := 0.0;
Well_Temp_Factor AT %MD5 : REAL := 0.0;
Optimal_Valve_Opening AT %MD6 : REAL := 0.0;
Steam_Flow AT %MD7 : REAL := 0.0;
Wellhead_Flow AT %MD8 : REAL := 0.0;
```

```
Wellhead_Pressure AT %MD9 : REAL := 0.0;
Wellhead_Temp AT %MD10 : REAL := 0.0;
Max_Wellhead_Pressure AT %MD11 : REAL := 2001.0;
Max_Wellhead_Flow AT %MD12 : REAL := 400.0;
Max_Wellhead_Temp AT %MD13 : REAL := 271.0;
Max_Separator_Pressure AT %MD14 : REAL := 2001.0;
Max_Separator_Temp AT %MD15 : REAL := 240.0;
Separator_Level AT %MD16 : REAL := 0.0;
Separator_Drain_Percent AT %MD17 : REAL := 0.0;
Separator_Drain AT %MD18 : REAL := 0.0;
Max_Separator_Level AT %MD19 : REAL := 1800.0;
Turbine_Exhaust_Temp AT %MD20 : REAL := 0.0;
Recent_Wellhead_Flow AT %MD21 : REAL := 0.0;
Wellhead_Override_Pressure AT %MD22 : REAL := 0.0;
MQTT_Wellhead_Pressure AT %MD23 : REAL := 0.0;
```

DNP3 VALUES

```
AnalogInputs:
Wellhead Temp - 0
Wellhead Pressure - 1
Wellhead Flow - 2
Steam Flow - 3
Separator Temp - 4
Turbine Drop - 5
Power Output - 6
Turbine RPM - 7
Turbine Output Temp - 8
Cooling Output Temp - 9
Reinjection Temp - 10
Separator Drain - 11

BinaryOutputStatus:
Dnp3_state - 0
```

SYSTEM ALERTS AND THRESHOLDS

Modbus Logic

If Wellhead Pressure, Wellhead Flow, or Wellhead Temp Over Max Trigger ESD

If StopSystem Then Wellhead Flow is set to 0

If Start System Then Separator Valve Opened to Optimal,

Steam Flow is then calculated based on Wellhead Flow, Separator Valve Percent and the Flash Factor

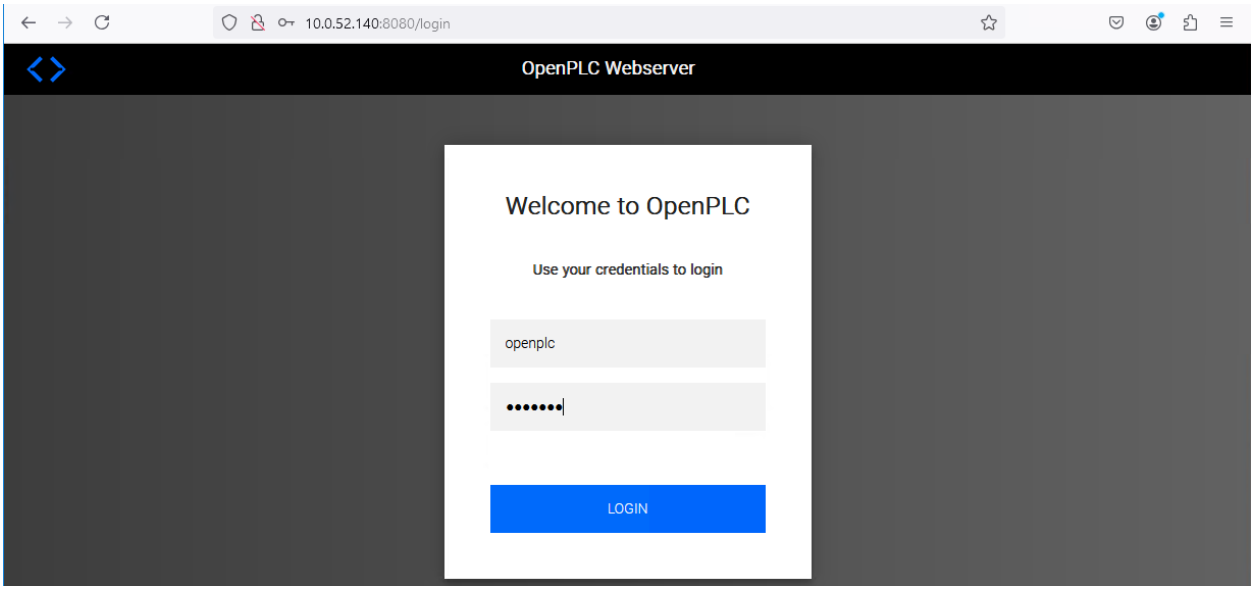
DNP3 Logic:

If Steam Flow > 0 and Turbine RPM is Not over Max, dnp3_state is True

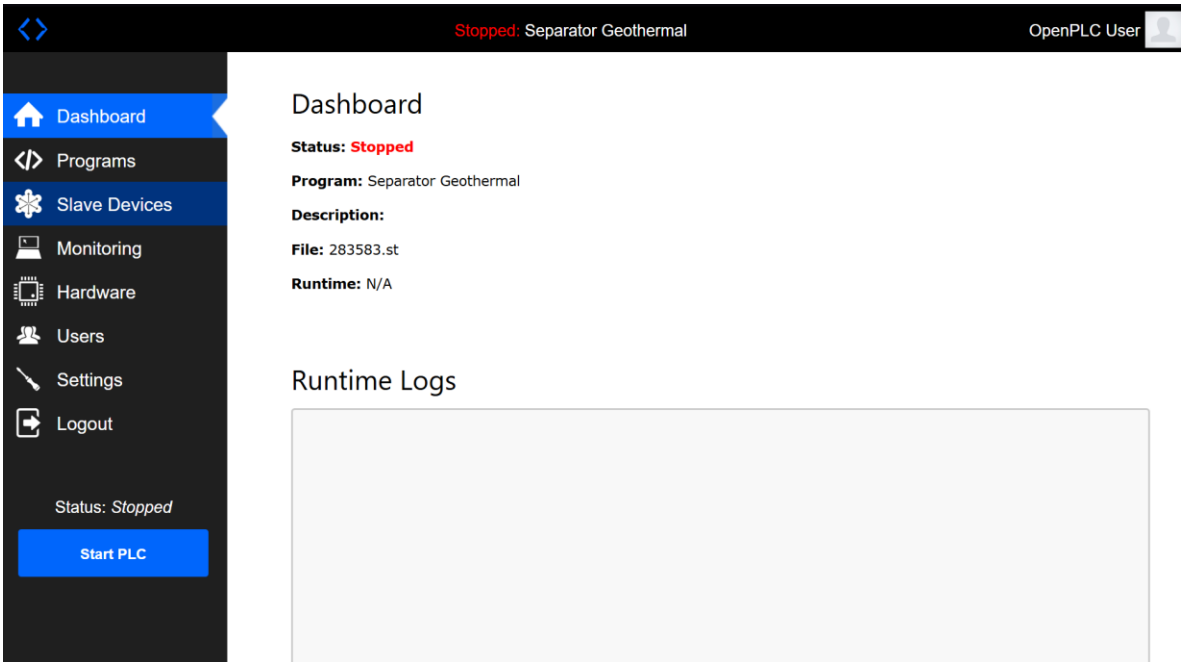
Stem Flow is then used to calculate Power Output.

PLC

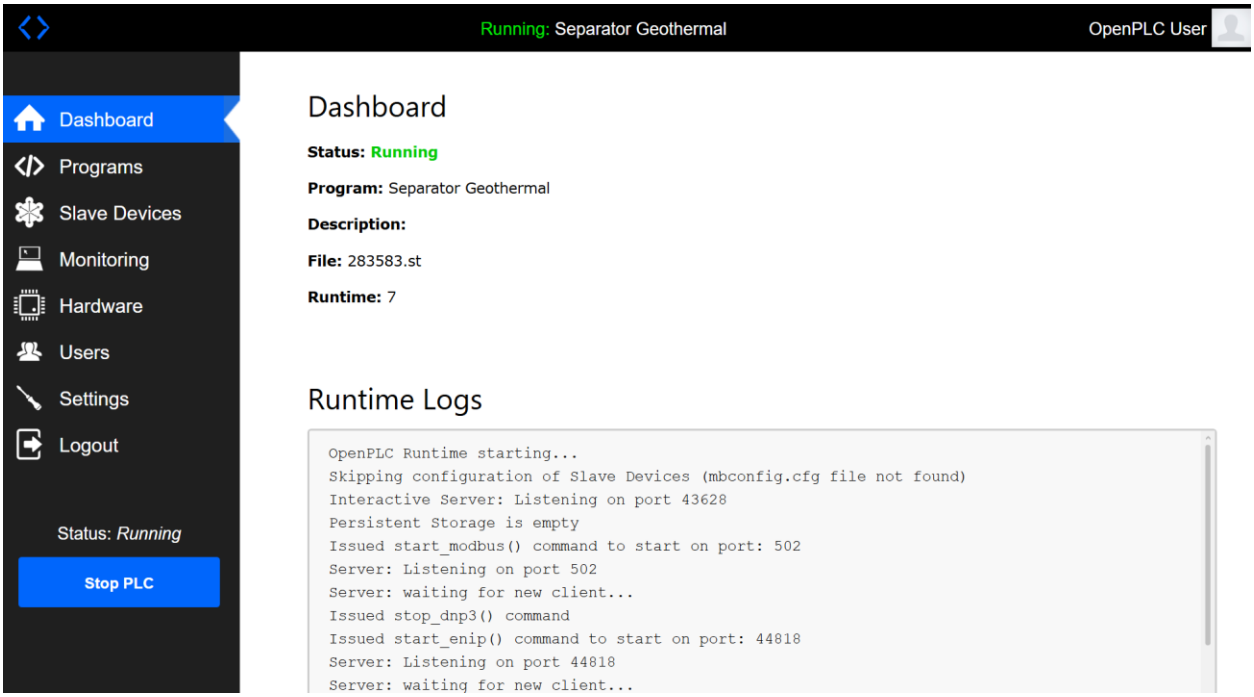
First, we must begin by starting the Modbus PLC, by going to <plc-ip>:8080. Here you will see the OpenPLC web page prompt. The credentials are openplc : openplc



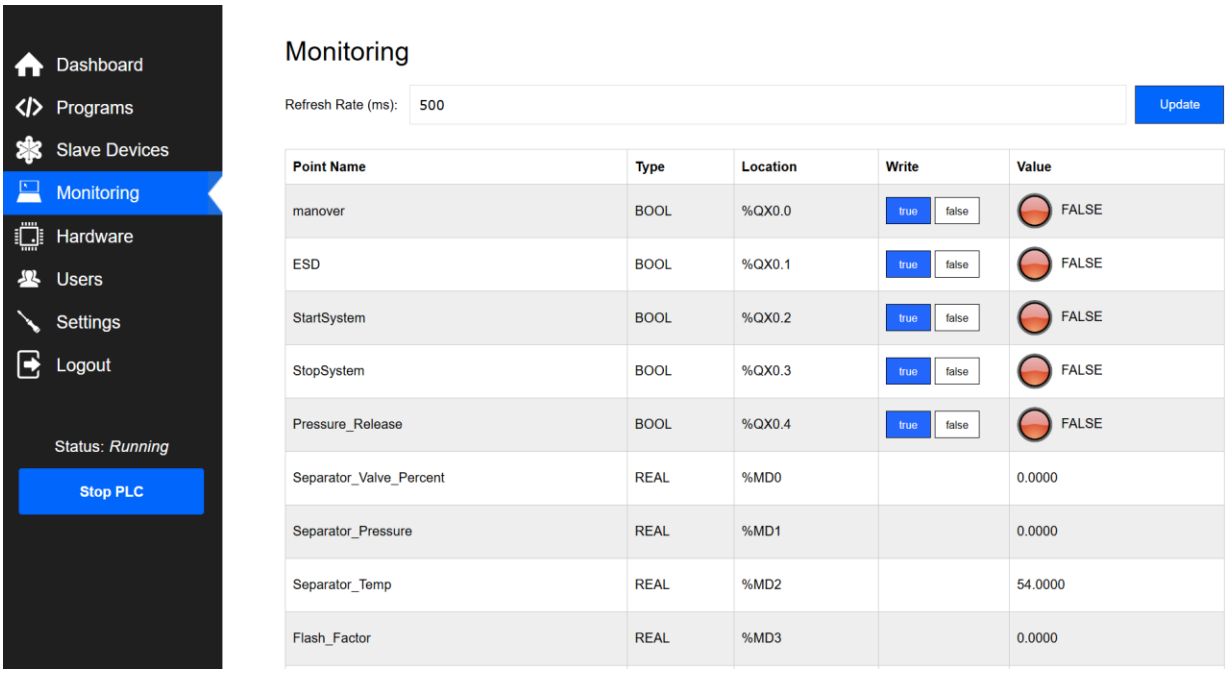
Once logged in you should see the dashboard, from here you can slick the blue button “Start PLC”. This will initialize the Separator Geothermal PLC Program.



Below is what you should see after the PLC program has started completely.



Below is the monitoring display where you can see all of the coils and registers associated with the PLC program along with their locations and current values.



Your DNP3 and MQTT connections will be enabled by default on your system. Both are running as services using Rust and should not be removed from your machines to prevent communication errors.

The DNP3 Service is located on the DNP3 PLC machine and its configuration and status are found at **`sudo systemctl status dnp3.service`**. It can be restarted with a similar command should for some reason during the competition it should stop using the command **`sudo systemctl restart dnp3.service`**

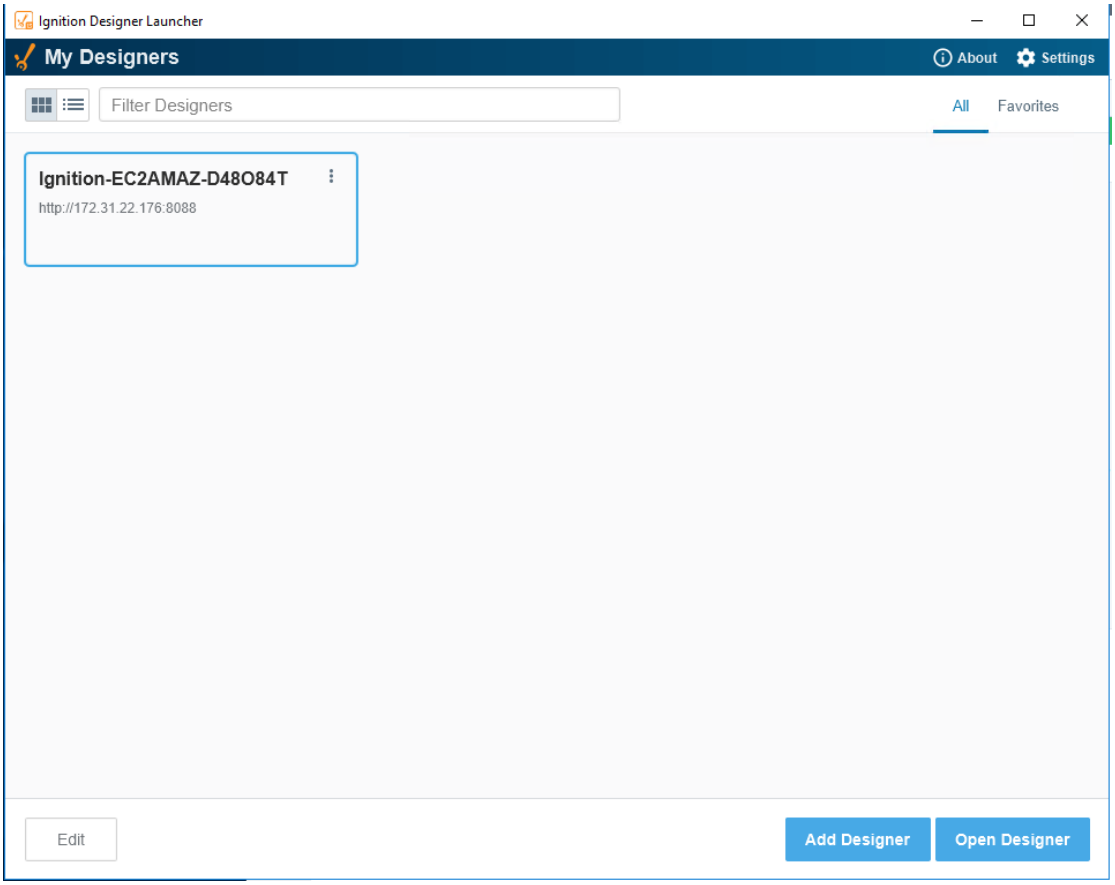
The MQTT communications from Modbus work the same way they are however located on the Modbus machine. This services status and configuration can be found here **`sudo systemctl status modbus.service.`** This can also be restarted using the following command **`sudo systemctl restart modbus.service`**

HMI

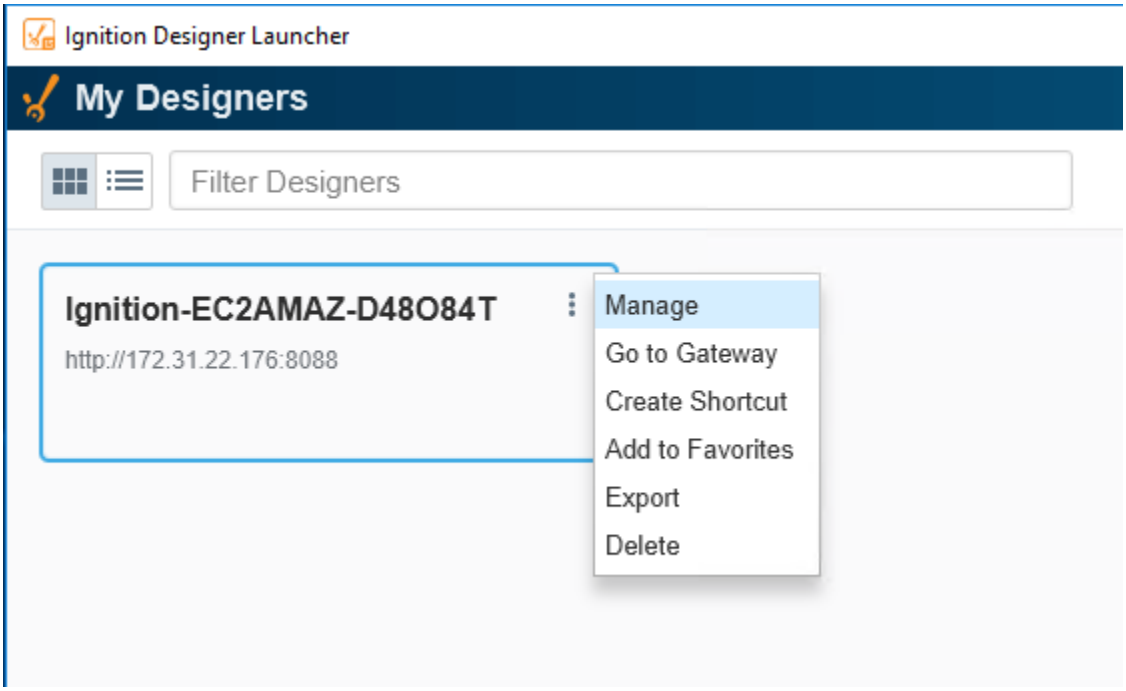


On the Desktop of the Windows 2019 CnC there are two icons designated for the Ignition HMI. Designer Launcher, launches the Ignition Designer application and the Perspective Workstation icon launches the HMI Perspective engineering workstation viewer.

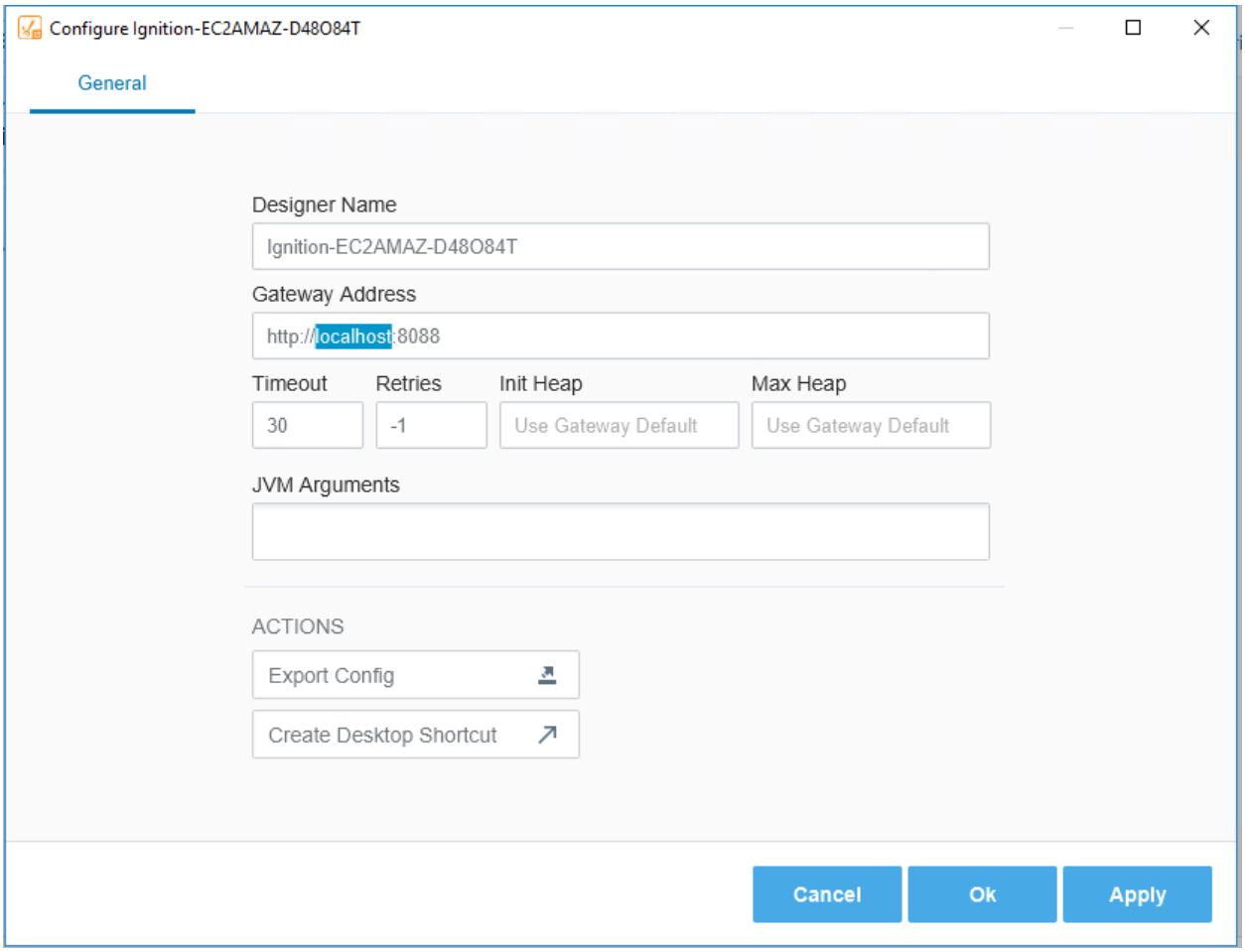
To ensure the Ignition Designer is properly connected, please follow the next steps.

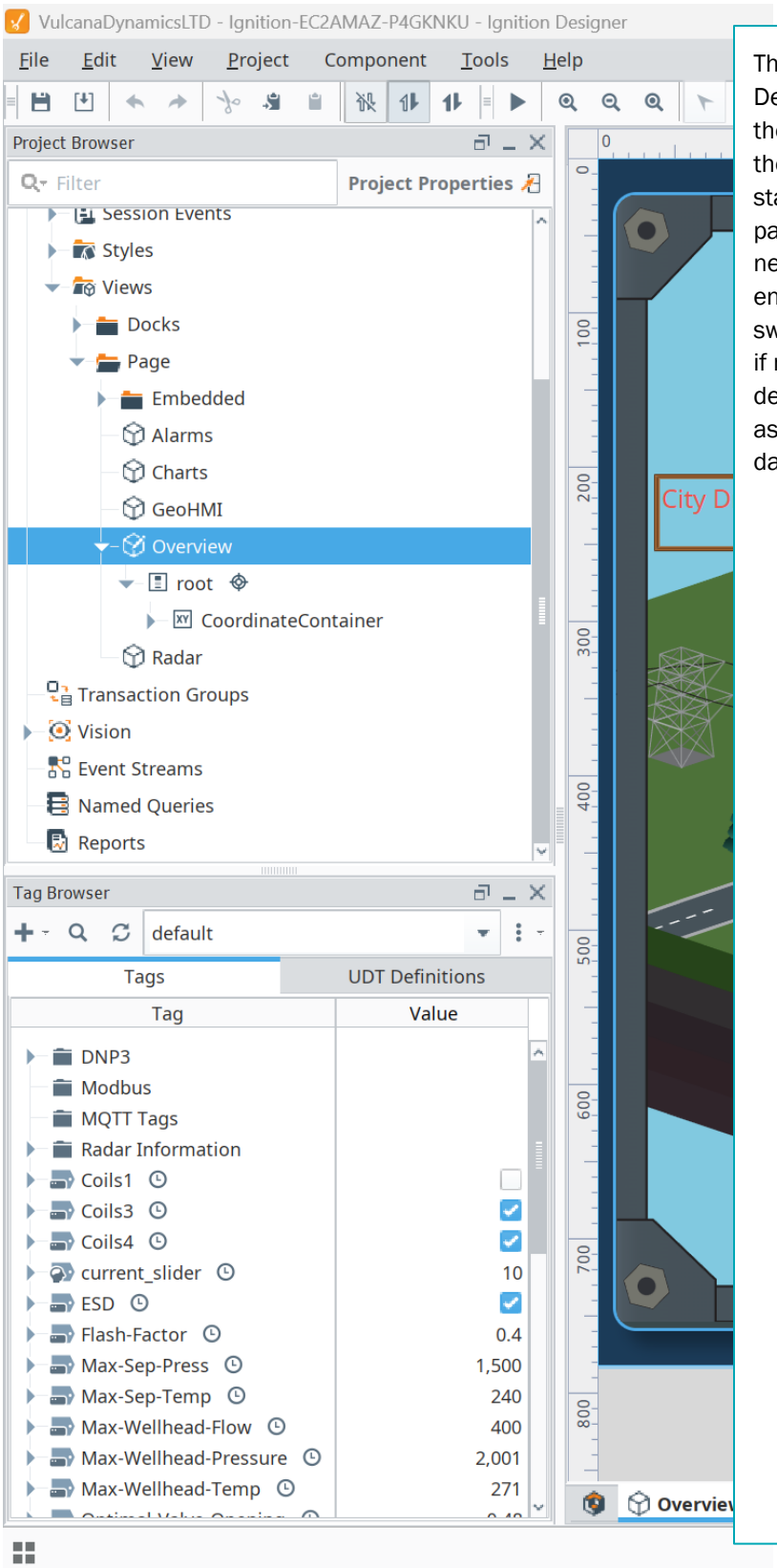


Click the vertical ellipses and then click “Manage” from the prompted window.



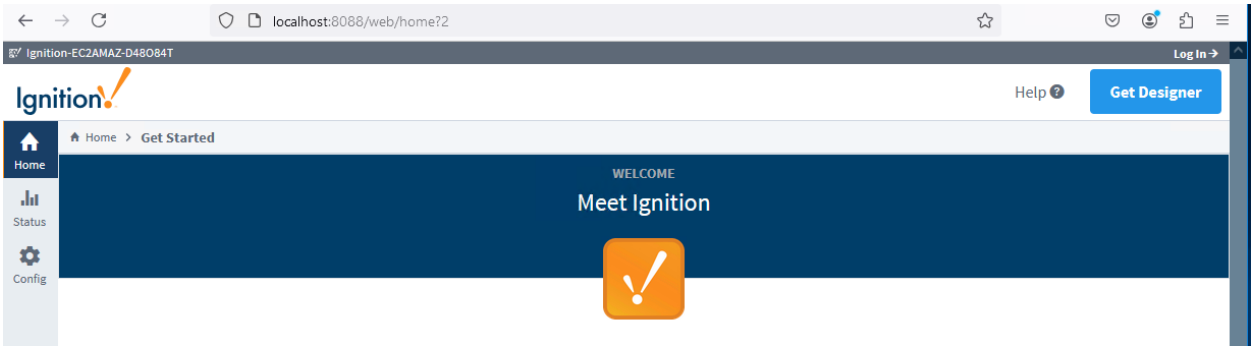
Edit the gateway address to reflect http://localhost:8088 Click the blue “Apply” button followed by the “Ok” button. Then click the “Open Designer” button to start the designer.



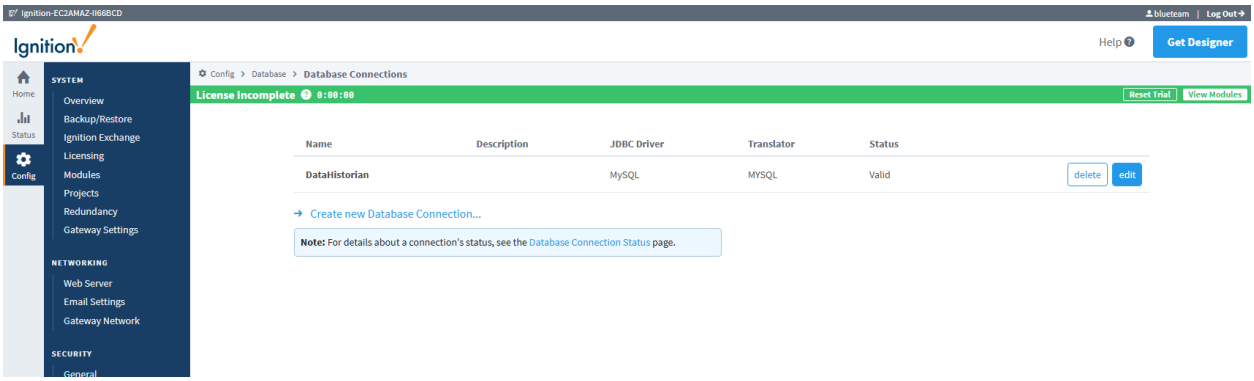


The image at the left shows the Ignition Designer application. Here is where all of the tag and database data can flow into the HMI views to indicate the current status', alarms, modbus data, etc. Each page view is constructed to show the necessary data to be shown to an engineer along with the appropriate switches and a manual override to utilize if necessary. Each tag corresponds to the designated modbus coil or register it is assigned, along with the corresponding database table attributes.

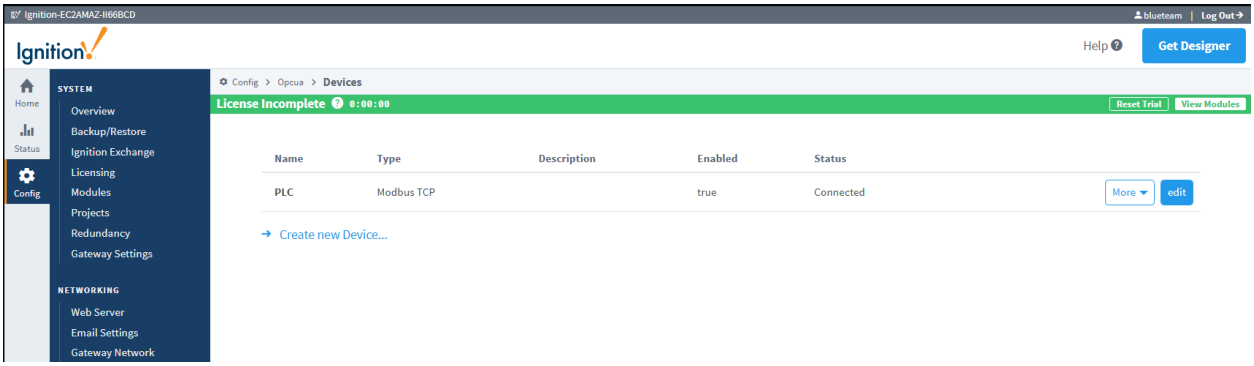
To access the Ignition Gateway, where all the driver connections are established, browser to localhost:8088 and the Ignition splash page will appear and allow login with blueteam: BlueTeam2026!



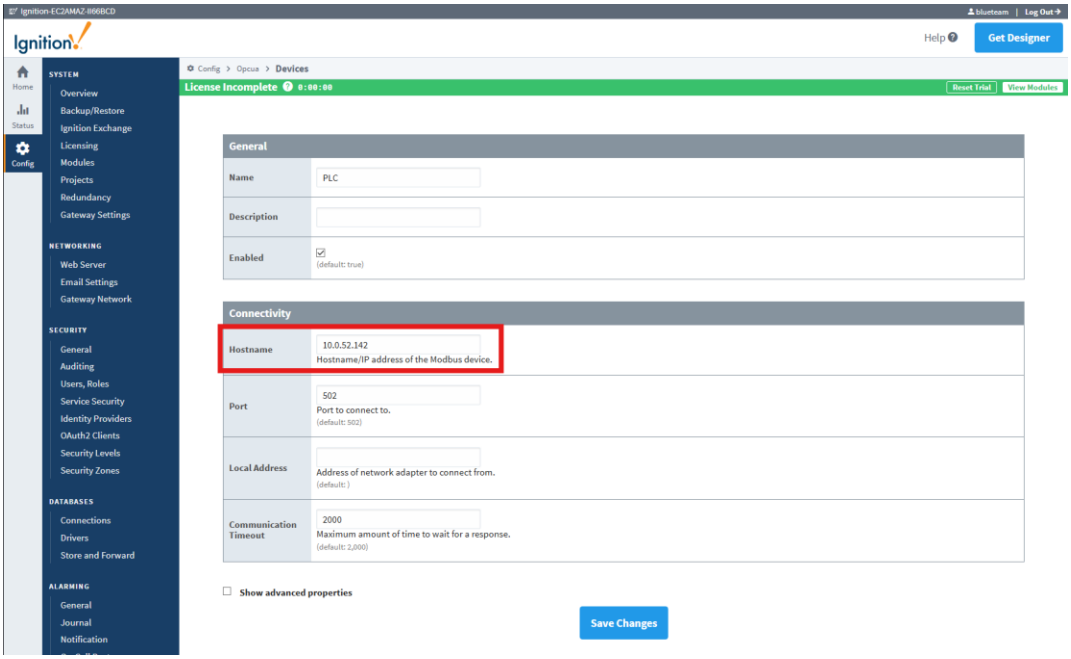
Once logged in, on the left-hand side, there are configuration tabs to drill down further into each type of connection for the HMI. The below image shows the database connection breakdown.



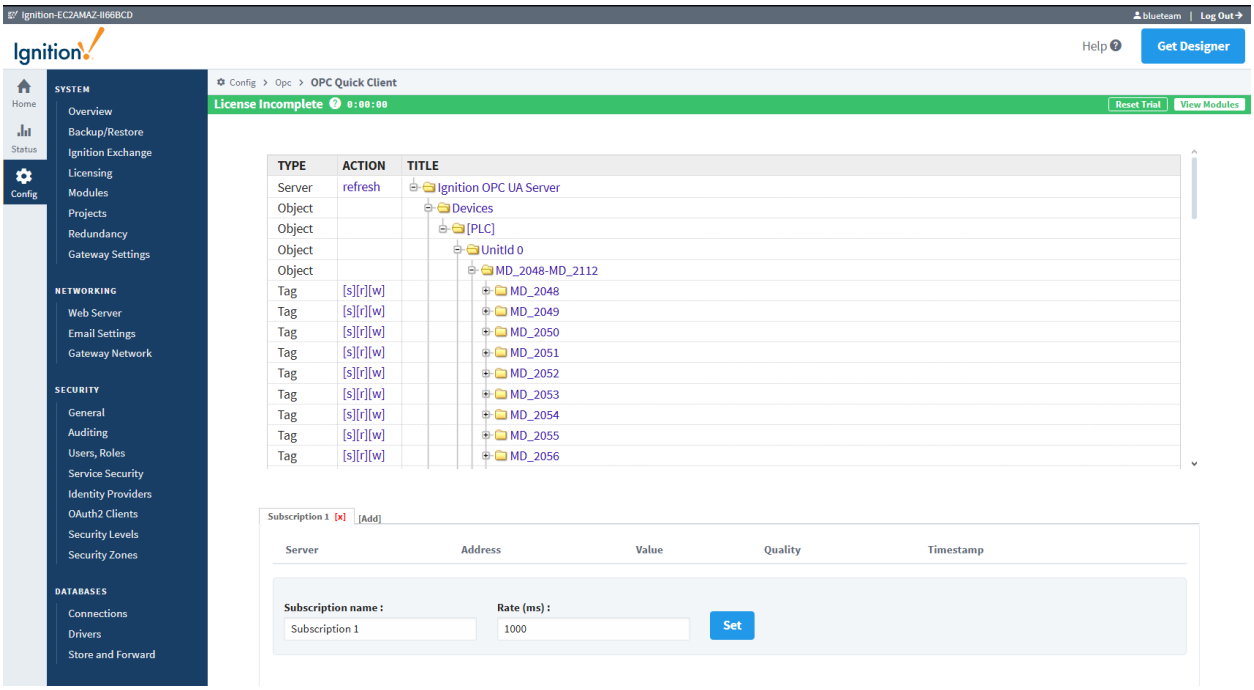
The image below shows the OPCUA connection breakdown for grabbing Modbus data from coils and registers.



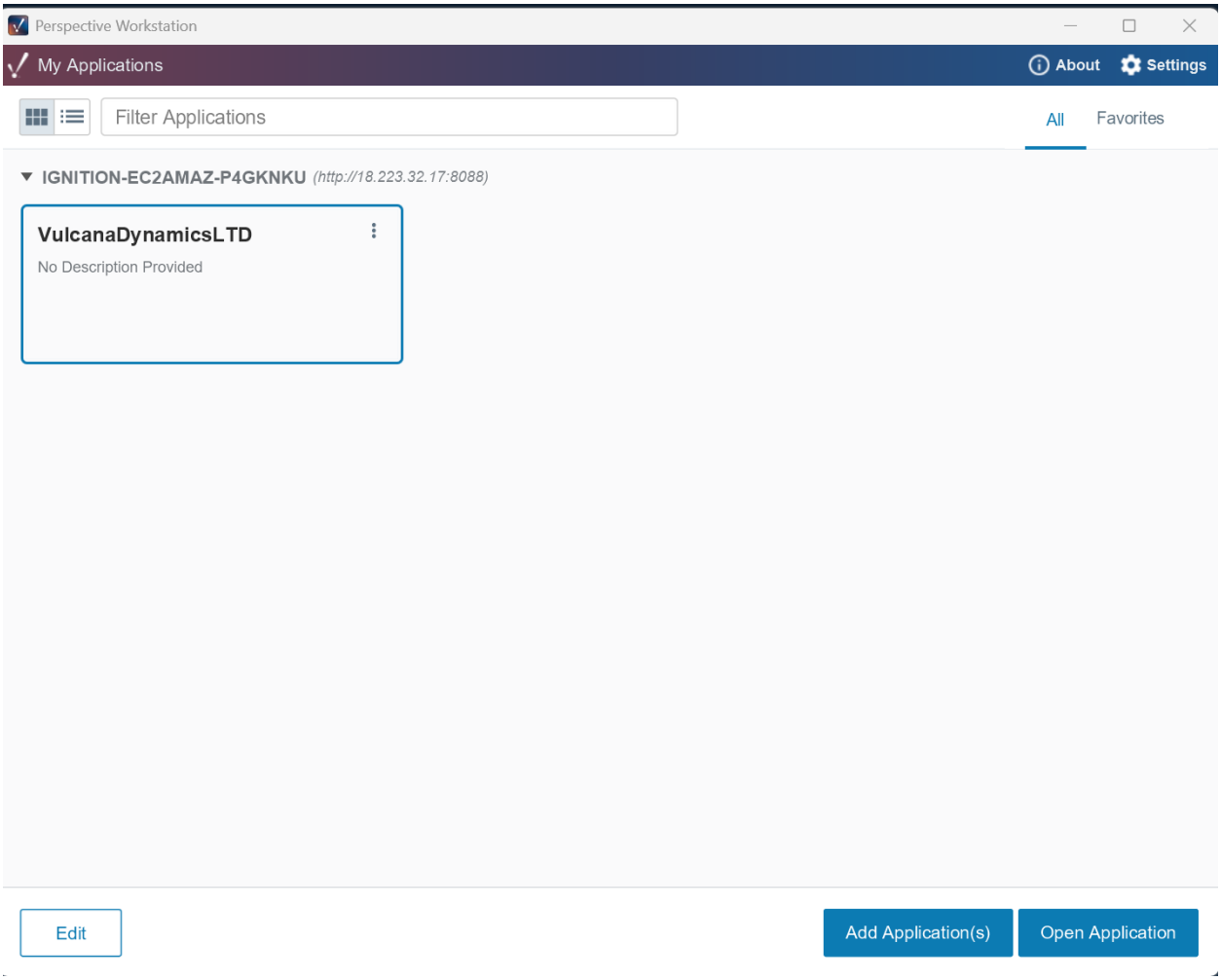
Upon Initially receiving your CNC virtual machine, you will need to edit the hostname of the PLC in OPC UA connection. To do this, you will click "edit" on “PLC” device found on the OPC UA Device connections page and update the Hostname field with the private IP of your PLC virtual machine.



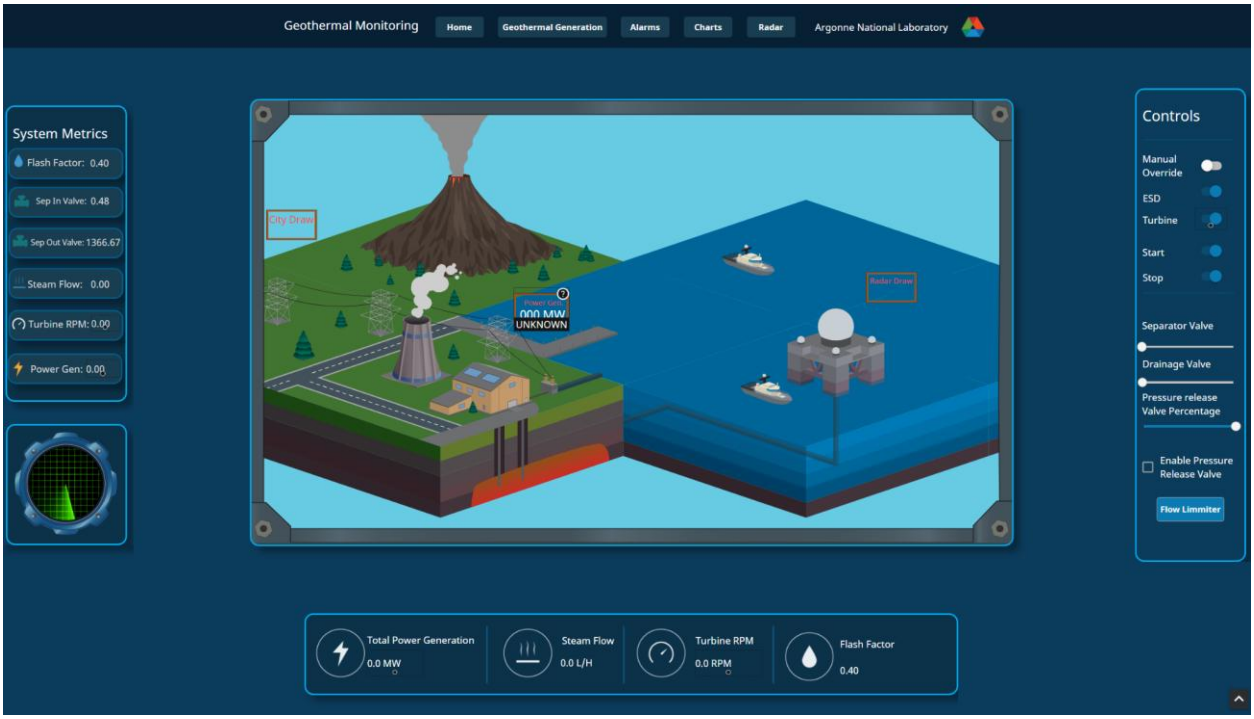
The following image is a further drill-down into each OPC quick client connection. This can show each read and write from/to modbus.



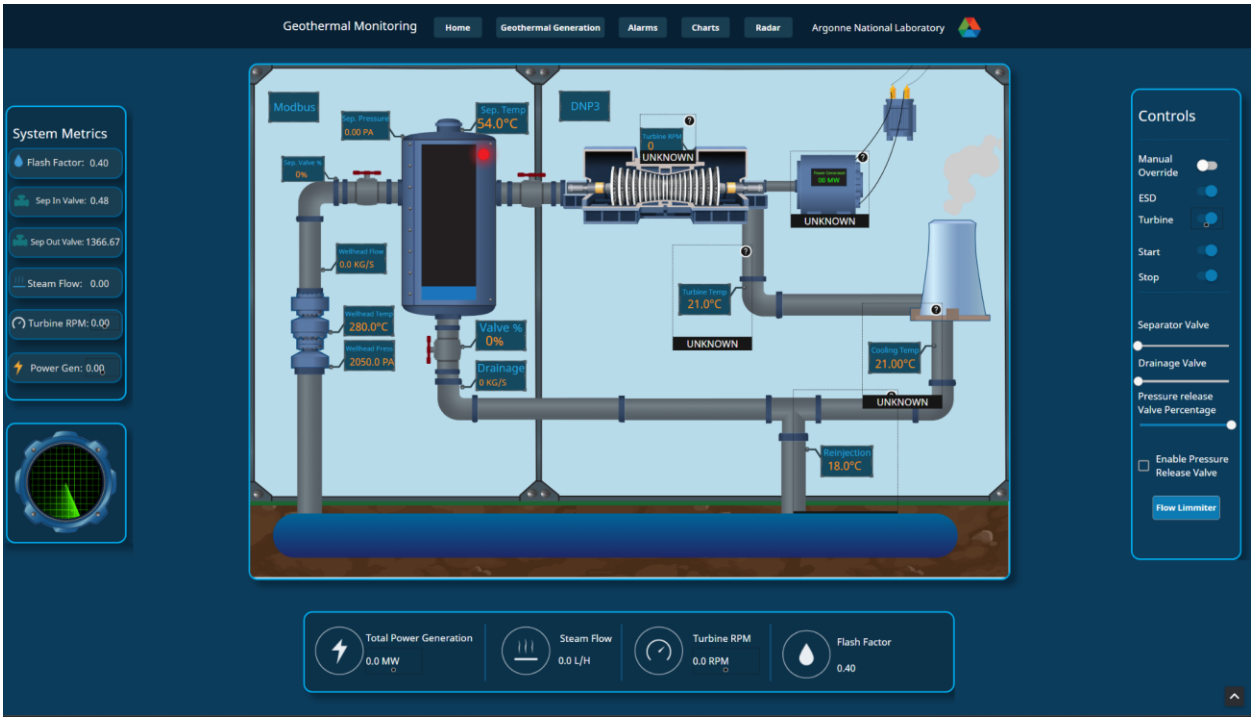
To start the Perspective Engineering Workstation view, click on the desktop icon to bring up the Perspective Workstation application. Click on the HMI application within and click the “Open Application” button. You will be prompted for login credentials.



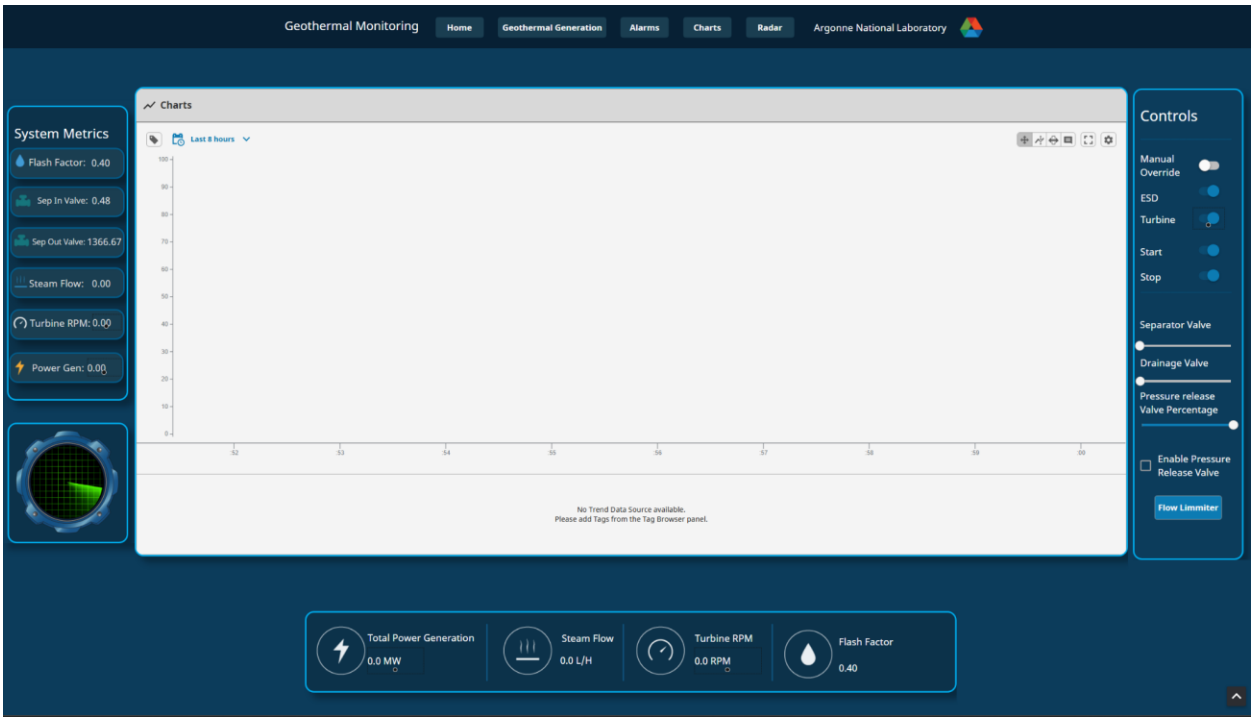




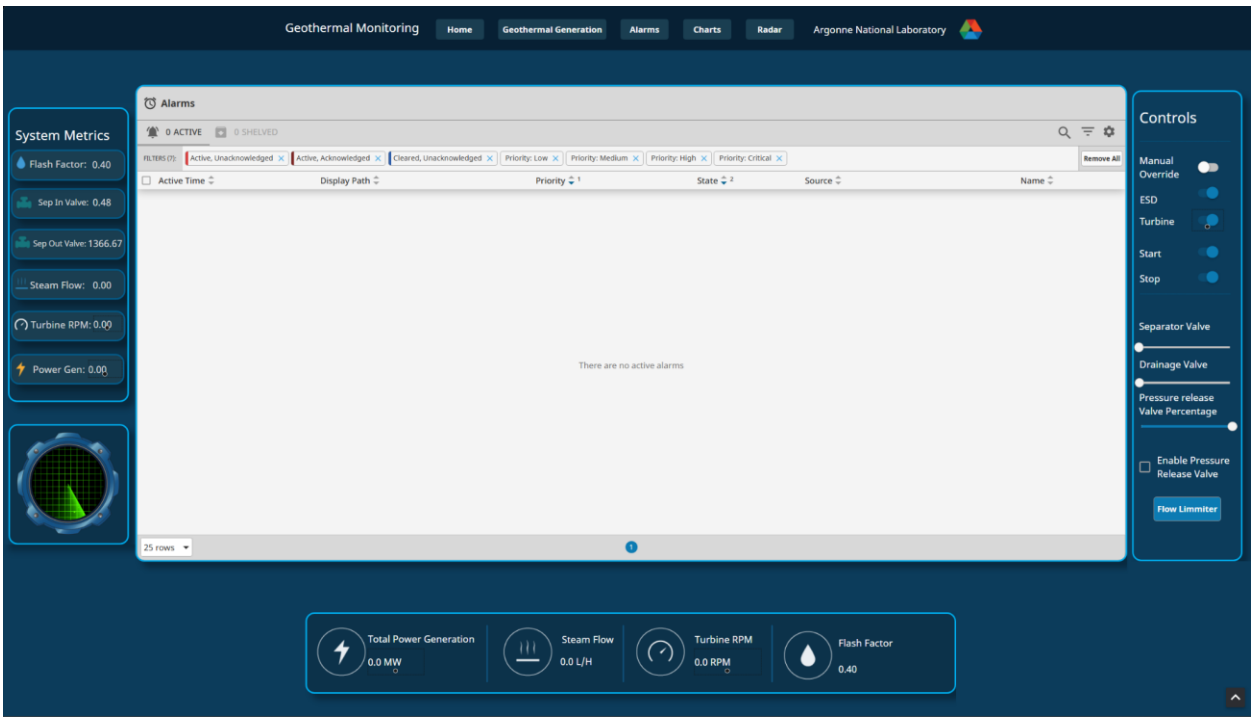
The Home page of the HMI will appear. This is where you can see the current system status.



The Geothermal Generation page is where you can see individual sensors.



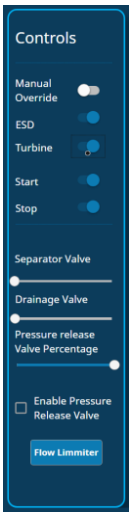
The Charts page directly corresponds to the output of the system. You can See graphical output over time, or production at specific timestamps.



The Alarms page shows all of the system alarms and provides the user the ability to acknowledge, shelf, or remove all alarms thrown by system threshold logic.

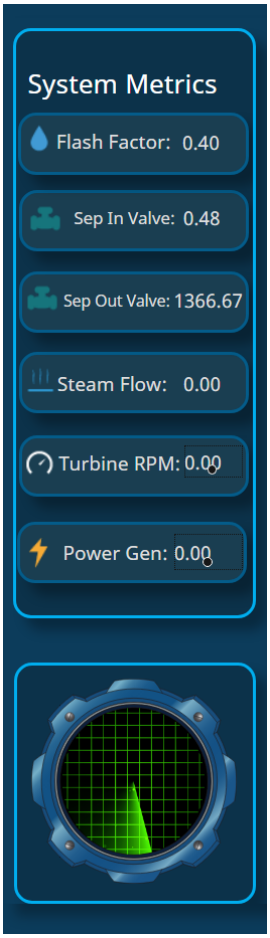


The Radar page helps to keep track of your current ICS score. By maintaining enough power to the Radar, you will be able to detect anomalous objects throughout the competition. It also shows you your overall uptime and what percentage of ICS points you have received out of the points available.



The Controls panel is available on every page of the HMI but is the location for all controls of the system. The Pressure release valve can be manually used, when the system shuts down due to overpressured wellhead.

- Manual Override - Using this allows for the system to be overridden outside of normal logic or safety parameters. This allows for manual control of various valves throughout the system.
- ESD - This triggers automatically when the system exits normal safety parameters to prevent damage and injury. It must be manually disabled after it is triggered. You must manually enable the start toggle to reinitialize the system from an ESD failure.
- Turbine - This will determine if the turbine is locked in place or not. When disabled the system will not generate power.
- Start - When triggered the system will go through the operations to start the system. The system will remain on as long as this is triggered.
- Stop - This is used to do a manual stop of the system. It takes precedence over the start switch and will force the system to go through normal stopping procedure.
- Separator Valve - This slider will change the percentage that the separator valve allows out of the separator tank and into the turbine. This will alter the amount of energy generated.
- Drainage Valve - This slider will change the percentage that the Drainage Valve is open/closed. Altering the amount drained from the separator into the return system.
- Pressure Release Valve Percentage - This slider will change the percentage that the Pressure Release Valve is open/closed.
- Enable Pressure Release Valve - Checking this box will enable the usage of the Pressure Release Valve percentage of open/closed with the Pressure Release Valve Percentage Slider.
- Flow Limiter (Button)- Directly lowers the amount of fluid from the wellhead into the separator system.



The Systems metrics panel is used to give the most important information about the system at all times. Here you will find the metrics that determine how much power your system is generating.

- Flash Factor - Calculated from the current pressure of the separator, this is the ratio of how much steam is generated from the water flowing into the system.
- Sep In Valve - How much fluid is currently flowing into the separator.
- Sep Out Valve - How much fluid is currently flowing out from the separator.
- Steam Flow - How much steam is currently being created and distributed to generate power.
- Turbine RPM - The speed at which the turbine is currently spinning in RPMs.
- Power Gen - How much power currently being generated from the system.

Below you can see the Vulcana database table structure breakdown. This table is used for the historical data on the ICS systems. It can be found on the HMI machine.
The credentials are **historian : CyberForce2026!** and should not be changed.

